

Perceptions on Corneal and Eye Donation: Implications for Corneal Donor Programs

Ian A. Pshea-Smith, MPH,* Mary Bashshur, BS,† Wadad Itani, MPH,‡ Zeinab (Noora) K. Mofrad, MPH,‡ Elizabeth Stout, MPH,‡ and Ramona Bashshur, JD, MJ§

Purpose: The purpose of this study was to review available literature that discusses the attitudes, awareness, and perceptions among individuals and populations regarding corneal and eye donation worldwide, to identify barriers that hinder donor programs from achieving sustainable success.

Methods: A search was performed in PubMed yielding 800 articles which resulted in a final set of 101 relevant articles for review. Articles were excluded after abstract review if the authors determined the articles did not sufficiently pertain to corneal donation. Information was extracted regarding year of publication, country of origin, income level, World Bank region, study aim, study setting, sample size, and percentage of consented donors. Qualitative information was extracted on barriers to corneal donation and discussions or mentions of eye banks.

Results: Of the 101 studies, 34% were in South Asia, 26% were in Europe and Central Asia, and 22% were in East Asia and Pacific. Fifty-six percent of the studies were published in upper-middle to high-income countries. Barriers to corneal donation were a common focus of the studies, and discussions of eye banks or eye banking systems appeared in 75% of the data set. Barriers to corneal donation in these studies were categorized as stemming from perceptions and awareness, family and religious factors, and societal factors.

Conclusions: The unique nature of eye donation emerged as a significant factor affecting willingness to donate. The interactions with trained donation coordinators are essential for the success of

eye banking service delivery, and eye banks remain the proven system for ensuring the availability of corneal tissue for transplant.

Key Words: eye donation, corneal donation, willingness, attitudes, perceptions

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Vision loss due to corneal blindness continues to be a global public health issue with distinct disparities across and within different regions and countries.^{1,2} Although estimates vary, it is known that outside of the United States, global need continues to far outpace the supply of donor corneal tissue and that transplantation remains the only scalable treatment of corneal blindness at the current time.¹⁻³ People who suffer from blindness and impaired vision confront numerous barriers to participation in society; thus, the work of vision restoration, which includes corneal transplantation, is key for eliminating hardships and increasing accessibility. Eye banking systems and corneal donor programs are the current proven, essential means of providing this cure.⁴ A shortage of tissues means that patients in need cannot access the treatment. This study aims to review available literature that discusses the attitudes, awareness, and perceptions among individuals and populations regarding corneal and eye donation worldwide, to identify barriers that hinder donor programs from achieving sustainable success.

METHODS

Search Strategy

We reviewed peer-reviewed studies which investigate perceptions and attitudes toward corneal donation. The initial search was performed in the PubMed database using the following search terms: ((“perception*” OR “understanding*” OR “behavior*” OR “attitude*” OR “willing*” OR “barriers*” AND (“donation*” OR “transplant*”) AND (“cornea*” OR “eyes*”). The search was limited to articles published in the last 20 years, from May 31, 2003, to May 31, 2023, returning 800 total initial results. The PubMed database was chosen for its comprehensive coverage of relevant, peer-reviewed literature.

Study Selection

All 800 abstracts were reviewed from the articles obtained in the initial search. Abstracts were evaluated

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From the *Department of Epidemiology, School of Public Health, University of Michigan, Ann Arbor, MI; †Department of Biology, College of Literature, Science and Arts, University of Michigan, Ann Arbor, MI; ‡Department of Health Behavior and Health Education, School of Public Health, University of Michigan, Ann Arbor, MI; and §Eversight, Ann Arbor, MI.

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Correspondence: Ramona Bashshur, JD, MJ, Eversight, Ann Arbor, MI 48108 (e-mail: rbashshur@eversightvision.org).

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independently by 2 authors for relevance to the research question and purpose of this review and were excluded based on mutual consensus. Articles were excluded if both authors concluded the abstract did not sufficiently pertain to perceptions and attitudes toward corneal donation. We did not exclude studies with small sample sizes as many of these were in-depth qualitative studies and were typically designed to obtain detailed information from participants regarding a highly personal topic.

Following the abstract review, 137 full-text articles were identified as potentially relevant to the study. These full-text articles were then retrieved and subjected to a comprehensive review. Following this review, we excluded 36 articles as follows: 17 articles were focused on all organs/tissues, including cornea but with no focus or discussion related to cornea. Five studies were letters to the editor, commentaries, or a magazine article. Four studies were focused solely on obtaining corneal donations/pledges for research registries and did not contain relevant discussion. We were unable to obtain full text for 4 studies. Three studies had a pure clinical focus such as new therapies or corneal tissue preservation time. One study was focused on the practices of the eye care sector related to consent for exported tissue only. One study was limited to analyzing the effects of COVID-19 on eye banking, and one study was concerned specifically with a hierarchy of donated organs and tissues. This resulted in 101 total studies for inclusion in this review (Supplemental References, Supplemental Digital Content 1, <http://links.lww.com/CB9/A14>).

Data Extraction

Information was extracted from each remaining study by one author with independent review by at least one other author, to ensure quality and consistency of results. The following variables were examined: year of publication, country of origin (including country income level and World Bank region), study aim, study setting, sample size, and percentage of consented donors. Qualitative information was extracted on barriers to corneal donation and discussions or mentions of eye banks or eye bank systems. All data extraction and analyses were performed using Microsoft Excel.

RESULTS

The 101 studies in our data set all investigated the knowledge, perceptions, awareness, or attitudes either of individuals or of their target populations. Table 1 contains the summary characteristics of included studies. Studies were concentrated in 3 regions: South Asia, Europe and Central Asia, and East Asia and Pacific. The 22 studies from the East Asia/Pacific region were all conducted in China while 25 of the 34 studies in the South Asia region were conducted in India, which has the world’s greatest burden of corneal blindness.⁵ Europe and Central Asia comprised 26 of the studies, with most of these conducted in the United Kingdom and Germany. A full 85% of our studies were published on or after 2010, with only 16% having been published in the time

TABLE 1. Summary Characteristics of Included Studies (N = 101)

Study Characteristic	Frequency	Percent
Region		
East Asia and Pacific	22	22
Europe and Central Asia	26	26
Latin America and Caribbean	1	1
Global	2	2
Middle East and North Africa	6	6
North America	3	3
South Asia	34	34
Sub-Saharan Africa	7	7
Country income classification		
High income	40	40
Upper-middle income	16	16
Lower-middle income	39	39
Low income	4	4
Year of publication		
2000–2009	16	16
2010–2019	48	48
2020–2023	37	37
Study aims		
Factors and attributes	78	78
End-of-life care	6	6
Program effectiveness	12	12
Other	5	5
Study setting		
Clinical	33	33
Eye bank or eye bank society	13	13
Education	19	19
General population	21	21
Internet-based	4	4
Other	11	11

period from 2000 through 2009. A majority of the studies (55%) were published in upper-middle to high-income countries, with low-income countries comprising only 4% of the studies. While all the articles discussed attitudes, knowledge, and awareness of corneal donation to some degree, 12% were concerned with evaluations of existing programs and 6% were narrowly focused on the end-of-life experience. The remaining 78% focused exclusively on a population’s perceptions relating to these key points. The study setting varied considerably, with the largest category (33%) having been conducted in a clinical setting, another 21% in general populations or communities, 19% in educational settings, and only 13% in an eye bank setting (Table 1).

Barriers to donation and consent were a common theme across the studies. Figure 1 illustrates these barriers as belonging to one of three thematic domains: 1) individual awareness, knowledge, and attitudes; 2) religious factors; and 3) societal factors.

DISCUSSION

This dataset comprised 101 unique studies focused principally on understanding the knowledge, perceptions,

Thematic summary of barriers to corneal donation		
Awareness, Knowledge, Attitudes	Religious factors	Societal Factors
Personal perceptions among potential donors and/or their next of kin (NOK)	Personal beliefs regarding practice of one's religion	Societal level awareness through education or training
Family objection or refusal from NOK, with or without stated reasonings	Influence of family members regarding religious practices and customs	Trust in health systems /other systems
Potential donor not wishing to upset family members	Nonspecific citation of religious barriers	
Perceived unsuitability due to health conditions		

FIGURE 1. Thematic summary of barriers to corneal donation.

awareness, and attitudes of individuals or specific populations worldwide toward corneal donation. Most of the studies reported that a significant proportion of the respondents indicated a willingness to donate, which must be regarded as, if not a prerequisite to donation, at least an indication that there are far more likely corneal donors in a given population whose family members are not approached. Important findings relating to repeated themes across the studies were found, with distinct relevance for the global eye banking sector and for global public health as a whole. The increase of studies in recent years indicates that greater attention is being focused on the worldwide shortage of donor corneal tissue. While high concentrations of studies focused on India and China, which is expected, other locations such as Africa are distinctly underrepresented in this data set. This is an issue of global health equity. Furthermore, the barriers to corneal donation in distinct geographical contexts may differ, and it may be harmful to generalize the findings from one location onto another.

The studies provide a diverse body of literature describing the challenges involved with the establishment of consented donor programs. Along with the barriers to corneal donation, motivations to donate or identification of who is more likely to become a donor were also discussed. While the motivations to donate can predominantly be conceptualized as variations on a theme of basic human altruism, there were several distinct patterns among donors and people who were more likely to become donors. Education and involvement in the health care field were strong factors of personal willingness. Although a certain level of education and awareness is a desirable precursor to establishing a donation program, general public awareness of corneal donation did not necessarily correlate with actual donations.

The importance of the end-of-life experience and the process for corneal donation should not be underemphasized. Corneal donation does not implicate the ethical debates regarding when life ends; thus, it can be somewhat counter-intuitive when people are willing to donate organs, either their own or family, but not eyes. Rather, there is a unique

dimension, often discussed in these studies, regarding how people feel about either their own eyes or loved ones' eyes. The concept of disfigurement was featured in these studies. This particular fear can exist on a personal level⁶ and is also reflected in the attachment and connection that people have for each other through a person's eyes. The phenomenon of eyes as "the windows to the soul" as a strong factor underlying family refusal to donate eyes even if organs were donated.⁷ It is compelling to note the articulated reactions of family members regarding "selective donation" that excluded eyes; as one respondent stated, "I've looked into her eyes...I haven't felt her internal organs."⁸ These attitudes toward their loved ones' eyes appear across countries, religions, and cultures.^{9,10} Barriers to eye donation were discussed using a variety of methods, from qualitative descriptions to advanced quantitative analyses. Often multiple barriers to donation were mentioned, largely from the perspective of how to improve rates of corneal donation. Lack of trust in health care systems or fear of corrupt systems was also discussed.^{11,12} This particular barrier, that of trust, is not easily addressed, especially when populations are aware of historical abuses or have had personal negative experiences. Restoring trust will be essential and requires actions from all levels of society and governance. Where eye banking systems are established and become trusted partners in their communities, success stories of restored vision help overcome legacies of mistrust and fear.

The US system has been successful in meeting needs, in part because there is less need to address, but mostly because of the successes and best practices established by the US eye banking infrastructure. The history of eye and organ donation in the United States demonstrates how this infrastructure can be created and maintained, in the face of inevitable obstacles, including trust, which will exist in any population. Notably, the Minority Organ Tissue Transplant Education Program (MOTTEP) became enormously successful after confronting the essential issues behind the lack of African American/Black American organ donorship in the face of greater need of lifesaving transplants on behalf of this community.¹³ These

essential issues had to do with gaining credibility and demonstrating a commitment to minority communities that were asked to place their trust in a health system that oftentimes had failed them. MOTTEP adapted a two-pronged approach that was instrumental in gaining this population-level trust through its mission to safeguard and promote public health and prevention.¹⁴ In the same way, eye bank programs in areas of great need must overcome historical and interpersonal factors that prevent donation programs from taking hold and sustaining any progress. Since there is a demonstrated and continued difference in how potential donors feel about eye donation, these specific barriers which have been effectively addressed in the US eye banking system must also be addressed in any target regions, countries, or communities where the work remains to be accomplished. Notably, even in the United States, the shortage of donor organs is felt disproportionately by African Americans and Hispanics, and by extension, we can conclude this may apply to other minority groups as well. Trust in systems will be a continued obstacle and, if not addressed, can even continue to erode over time. The role of eye banks in creating this sense of trust cannot realistically be separated from the processes and components of corneal donation.

Another common issue was that families of the deceased may refuse to give consent and potential donors may not communicate their wishes while alive, for fear of upsetting family members. Researchers also identified that the timing of donation discussions may play a role in the effectiveness of approach.^{15,16} Health care workers in a variety of settings were found in some studies to have low knowledge levels regarding corneal disease, treatments, and the existence of eye banks and described a need for more specific training.¹⁶ Being part of the health care workforce did not necessarily confer any sense of ease for purposes of family discussions, particularly for providers who were not trained to approach families for consent to donate.¹⁷ Notably, even with specific training on approach and consent, the approach process was still not comfortable for these health care workers.¹⁸ Furthermore, while being a part of the health care workforce factors into end-of-life decision making, these individuals being more likely to donate themselves did not mean they were any more likely to be comfortable introducing donation discussions with families.¹⁹

Rulings and guidelines from religious authorities at higher levels are necessary and have been issued by most major world religions and other smaller denominations; however, similar to the education component, this is only the first step. Researchers discussed or even measured religion as a factor, yet it is not possible to draw conclusions about the impact of any particular religion or even of religion in general. However, we would emphasize it is likely that the end-of-life experiences override any such possible trends identified in the literature. The personal nature of these experiences may not necessarily lend themselves to quantitative analyses, and certainly, survivors and family members are the essential key to what happens regarding donations after life is over. The complexity of religious influences can result in confusion and more reluctance to donate. Of course, the decedent's involvement in religion may aid in family

discussions, and relationships with local religious leaders at the family level are known to be influential in decision making. The family unit and its connectedness vary among different regions and culture, and the relevant dynamic can result in circumstances that lead to a refusal to donate eyes even if the decedent had wished to donate. Understanding the importance of this family dynamic, and how it intersects with other factors, is crucial for tailored donation programs and requires a level of dexterity on the part of the coordinators who will encounter this variation of characteristics within their communities.

As eye banking systems are the required operational infrastructure for the donation and transplantation process, we were interested to see how eye banks were represented in these studies. There were 3 distinct categories; in the first group, either the role of eye banks and related infrastructure was discussed in detail, the eye bank was the study setting and/or part of the research group, or there was other significant discussion of the role of eye banking. These comprised 59% of studies. In the second group, 17% of the studies, eye banks were implicitly or even explicitly acknowledged, but there was not extensive discussion of the role of eye banking or its functions. For some of these studies, especially those that used questionnaires, the researchers included questions designed to measure participant awareness of eye banks and perhaps, additional details regarding the donation process. Thus, most of the studies (75%) either had extensive discussion of the eye banking role or at least had an explicit or implicit mention of the eye banking infrastructure. The remaining studies (25%) did not mention eye banking or eye bank systems.

With a majority of studies containing some element of discussion regarding eye banking or eye banking infrastructure, the 25% that did not mention eye banks may offer some insights into how partnerships may aid the progress of new eye bank developments in areas where they are getting started or are seeking greater sustainability, for example, in palliative care settings. In particular, a few of the less recently published studies examined personal experiences in the end of life that offered rich context for understanding.²⁰ A few studies offered a detailed analysis of some portion of the eye banking model without identifying it as such, for example, the role of the dedicated donation coordinator, a position that holds the primary potential for increasing donations.²¹

This role was demonstrated to be of primary importance whether described as a formal job function, such as the eye donation coordinator (EDC),²² or as a less specific means of initiating the conversation.²³ One study concluded that experience and training were more essential for obtaining donations, than whether the approach was in person or by telephone.²⁴ All the described barriers to eye donation are confronted by the trained coordinators whose role is to speak with the families and medical personnel.²⁵ This discussion of eye donation is ideally treated as a related yet separate component from the concept of organ donation. Because of peoples' unique perspective regarding their eyes, and their loved ones' eyes, the training of the donation coordinators involves a unique set of skills and is not necessarily analogous to the organ donation approach. The clear

impression from most of the studies is that it is these specific interactions that take place between a caring individual with the requisite training that forms the most essential component to the realization of the gift of corneal tissue.

It is also important to recognize that for some people there exists a personal rejection of the donation concept that is inflexible, and thus, donation will always remain an unacceptable health care concept for a subset of any population. This group of people who are not potential donors should know that their wishes will be respected as well; such a non-discriminatory application of respect can help overcome more general lack of trust. Given these different categories of obstacles to be overcome in gaining consent and the need to respect refusal, here again the role of the trained donation coordinator is paramount and encompasses all of these.

CONCLUSIONS

A substantial body of literature, as shown in this study, has examined the impacts of differing cultural factors on donation decisions both during a person's lifetime and among their next of kin after they have died. These studies yield valuable insight into what has been achieved and what has yet to be accomplished. We conclude that the unique perceptions that people hold dear regarding their eyes, and the eyes of loved ones, inform all other aspects of donation. Thus, the importance of the contact between these surviving family and friends, and the people coordinating the donation process, cannot be overstated. Of course, community and public knowledge about the specifics of eye donation also has significance because many misconceptions can be addressed with accurate information. Eye banking infrastructure is a comprehensive and specialized system for corneal treatment within the health care system and larger public health infrastructure. The donation process for corneal tissue can and should be distinguished from the process for donation of organs and other tissues. There is no existing shortcut or substitute; alternative therapies are still not likely to be widely available for the foreseeable future. It is the existence of eye banking systems that is required to overcome all identified barriers to eye donation and thus address the ongoing shortage of corneal tissue, and it is this infrastructure that must be implemented in countries and regions with untreated corneal disease, in order that people living in these areas may also receive the lifesaving gift of sight.

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REFERENCES

1. Solebo AL, Teoh L, Rahi J. Epidemiology of blindness in children. *Arch Dis Child*. 2017;102:853–857.
2. Porth JM, Deiotte E, Dunn M, et al. A review of the literature on the global epidemiology of corneal blindness. *Cornea*. 2019;38:1602–1609.
3. Wang EY, Kong X, Wolle M, et al. Global trends in blindness and vision impairment resulting from corneal opacity 1984–2020: a meta-analysis. *Ophthalmology*. 2023;130:863–871.
4. Oliva MS, Schottman T, Gulati M. Turning the tide of corneal blindness. *Indian J Ophthalmol*. 2012;60:423–427.
5. Anitha V, Tandon R, Shah SG, et al. Corneal blindness and eye banking: current strategies and best practices. *Indian J Ophthalmol*. 2023;71:3142–3148.
6. Lawlor M, Kerridge I, Ankeny R, et al. Specific unwillingness to donate eyes: the impact of disfigurement, knowledge and procurement on corneal donation. *Am J Transpl*. 2010;10:657–663.
7. Bracher M, Madi-Segwagwe BC, Winstanley E, et al. Family refusal of eye tissue donation from potential solid organ donors: a retrospective analysis of summary and free-text data from the UK National Health Service Blood and Transplant Services (NHS-BT) National Referral Centre (1 April 2014 to 31 March 2017). *BMJ Open*. 2021;11:e045250.
8. Lawlor M, Kerridge I. Understanding selective refusal of eye donation. Identity, beauty, and interpersonal relationships. *J Bioeth Inq*. 2014;11:57–64.
9. Haddad MF, Khabour OF, Alzoubi KH, et al. Public attitudes toward corneal donation in Northern Jordan. *Clin Ophthalmol*. 2018;12:1973–1980.
10. Carey I, Forbes K. The experiences of donor families in the hospice. *Palliat Med*. 2003;17:241–247.
11. Tsigkos D, Tzelepi A, Kopsini D, et al. Interactive online survey raises awareness about cornea donation. *BMJ Open Ophthalmol*. 2020;5:e000285.
12. Uhlig CE, Promesberger J, Hirschfeld G, et al. Results of an internet-based survey amongst members of the German Ophthalmological Society concerning postmortem cornea donation. *Ophthalmologie*. 2012;109:1198–1206.
13. *National MOTTEP*. Available at: <https://www.natlmottep.org/>. Accessed February 27, 2024
14. *About Dr. Clive Calendar*. Available at: <https://www.natlmottep.org/#AboutDrCliveCalendar>. Accessed February 27, 2024
15. Niday P, Painter C, Peak J, et al. Family and staff responses to a scripted introduction to tissue donation for hospice inpatients on admission. *Prog Transpl*. 2007;17:289–294.
16. Gillon S, Hurlow A, Rayment C, et al. Obstacles to corneal donation amongst hospice inpatients: a questionnaire survey of multi-disciplinary team member's attitudes, knowledge, practice and experience. *Palliat Med*. 2012;26:939–946.
17. Gopal BP, Titi-Lartey OA, Fernandes P, et al. Evaluation of junior doctors' knowledge of corneal donation and the new opt-out system in England. *Postgrad Med J*. 2022;98:e174–e176.
18. Stiel S, Salla S, Steinfeld A, et al. Evaluation of the need for communication training of ophthalmologists for gaining telephone consent for cornea donation. *Ophthalmologie*. 2011;108:151–155.
19. Giugni C, Cecchi C, Santucci C, et al. Why is corneal donation so rare in children's hospices? A survey of multidisciplinary team members attitudes, knowledge, practice, and experience. *Pediatr Transpl*. 2022;26:e14217.
20. Walker L, Neoh K, Gilkes H, et al. A qualitative study using semi-structured interviews of palliative care patients' views on corneal donation and the timing of its discussion. *Palliat Med*. 2018;32:1428–1437.
21. Sharma B, Shrivastava U, Kumar K, et al. Eye donation awareness and conversion rate in hospital cornea retrieval programme in a tertiary hospital of Central India. *J Clin Diagn Res*. 2017;11:NC12–NC15.
22. Panigrahi S, Mahapatra B, Pati S, et al. Obstacles to post-mortem cornea donation: a study from a tribal region in eastern India. *Cureus*. 2022;14:e25176.
23. Schmidt S, Ghouzi A, Petrov G, et al. Cornea donation in Germany: obtaining consent. *Clin Transpl*. 2020;34:e13895.
24. Geissler A, Gerbeaux PR, Maitrejean C, et al. Cornea donation: evaluation of a training session to obtain consent by telephone. *Transpl Proc*. 2005;37:4634–4636.
25. Kacheri A, Mudhol R, Chougule S, et al. Eye donation: awareness, knowledge, willingness, and barriers among paramedical and allied health science students at a tertiary care teaching hospital in South India. *ScientificWorldJournal*. 2022;2022:5206043.